Junbo Zhao

(86) 18343087267 | zhaojunbo2012@sina.cn

Personal Website: https://a-lincui.github.io/

EDUCATION

Tsinghua University, Beijing, China 09/2021-06/2024
Master's degree in Electronic and Information Engineering
Thesis Title: Automatic Design Method of Efficient Neural Network Based on Performance Prediction and Weight sharing

Tsinghua University, Beijing, China 09/2017-06/2021
Bachelor's degree in Engineering Physics

PUBLICATION

- **Junbo Zhao***, Xuefei Ning*, Enshu Liu, Binxin Ru, Zixuan Zhou, Tianchen Zhao, Chen Chen, Jiajin Zhang, Qingmin Liao, Yu Wang, "Dynamic Ensemble of Low-fidelity Experts: Mitigating NAS 'Cold-Start'," AAAI 2023, Vol. 37 No. 9: AAAI-23 Technical Tracks, pp. 11316-11326.
- Ning X*, Zhou Z*, **Zhao J**, et al. TA-GATES: An Encoding Scheme for Neural Network Architectures[J]. Advances in Neural Information Processing Systems, 2022, 35: 32325-32339.
- Ning X*, **Zhao J***, Li W, et al. Discovering Robust Convolutional Architecture at Targeted Capacity: A Multi-Shot Approach[J]. 2020.DOI:10.48550/arXiv.2012.11835.
- Gao Z, Liu S, **Zhao J**, et al. Ensemble-based Reliability Enhancement for Edge-Deployed CNNs in Few-shot Scenarios, in IEEE Transactions on Machine Learning in Communications and Networking, doi: 10.1109/TMLCN.2024.3435168.
- Yu Wang, Wenshuo Li, **Junbo Zhao**, et al., A Data Processing Method and Related Devices, Chinese Patent, (Under review).
- Yu Wang, Xuefei Ning, **Junbo Zhao**, Method, Device, Equipment, Storage Medium, and Program Product for Generating Text Adversarial Examples, Chinese Patent (Under review)

RESEARCH EXPERIENCES

Method, Device, Equipment, Storage Medium, and Program Product for Generating Text Adversarial Examples 10/2023-12/2023
Researcher Advisor: Yu Wang, Xuefei Ning

- **Adversarial Text Generation**: Developed a black-box adversarial text example generation scheme leveraging a fine-tuned large language model, improving generation capability and control.
- **Prompt Paradigm Design**: Designed a comprehensive prompt paradigm encompassing attack background, purpose, control, and input. Created a dataset of "prompt-effective adversarial examples" and fine-tuned a pre-trained large language model to enhance adversarial text generation.
- **Experimental Validation**: Conducted experiments on the SST-2 binary text sentiment classification dataset using the ChatGLM-6B model, demonstrating the effectiveness of the proposed method.

Verification Study of Low-Bit Neural Networks in Compute-in-Memory Inference Chips

09/2022-10/2023
Researcher Advisor: Yu Wang, Xuefei Ning

- **Fault-Tolerant Model Design**: Led research in designing efficient fault-tolerant neural network models for tasks such as super-resolution and object detection, incorporating joint quantization.
- **Predictor Development**: Developed an architecture performance predictor for fault-tolerant neural

architecture search and introduced a dynamic inference method to enhance fault tolerance.

AI Quantization Compression and AI Software-Hardware Architecture Design 09/2022-02/2023
Researcher Advisor: Yu Wang, Xuefei Ning

- **NAS for Super-Resolution**: Developed software for neural architecture search (NAS) tailored for super-resolution tasks, designing and validating the search space for UNET models.
- **Search Strategy Exploration**: Explored various search strategies, including differentiable search, reinforcement learning, and evolutionary algorithms, and wrote comprehensive feasibility reports.

Dynamic Ensemble of Low-fidelity Experts: Mitigating NAS "Cold-Start" 04/2022-08/2022
Researcher Advisor: Yu Wang, Xuefei Ning

- **Literature Review**: Reviewed existing research on predictor-based neural architecture search, highlighting the challenge of limited architecture-performance data.
- **Cold-Start Solution**: Proposed using low-fidelity information (e.g., one-shot estimation, zero-shot estimation) to mitigate the "cold-start" problem in predictor-based NAS. Developed a dynamic ensemble prediction framework to fuse beneficial information from different low-fidelity data types.
- **Paper Writing**: Authored a comprehensive research paper detailing the proposed methods and results.

TA-GATES: An Encoding Scheme for Neural Network Architectures 11/2021-01/2022
Researcher Advisor: Yu Wang, Xuefei Ning

- **Strategy Exploration**: Investigated existing strategies for encoding neural network architectures, with a focus on graph-based schemes like GATES.
- **Encoding Scheme Development**: Created TA-GATES, a novel graph-based architecture encoding scheme designed to simulate the training process of neural networks, improving the ranking of candidate architectures using limited data.
- **Validation**: Validated the efficacy of the proposed scheme in enhancing predictor performance and contributed to writing the research report.

Discovering Robust Convolutional Architecture at Targeted Capacity: A Multi-Shot Approach
09/2020-03/2021
Researcher Advisor: Yu Wang, Xuefei Ning

- **Robustness and Capacity Analysis**: Investigated the relationship between adversarial robustness in neural networks and model capacity.
- **NAS Methodology Improvement**: Identified limitations of the one-shot NAS method in discovering robust architectures and proposed training multiple super-networks with varying capacities to mitigate these issues.
- **Feasibility Validation**: Conducted experiments to validate the proposed strategy, exploring its main application scenarios.

PROFESSIONAL EXPERIENCES

Infinigence Technology Co., Ltd. 08/2023-12/2023
Algorithm Researcher

- **Model Efficiency Evaluation**: Evaluated the efficiency of large language models (e.g., GLM-6B, OPT-6.7B) on domestically produced chips (e.g., Tianyue, Haifeike) in China, providing crucial data for lightweight large language model design.
- **Chip Testing Solution**: Contributed to the design and development of a unified chip testing solution.

Novauto Technology Co., Ltd. 03/2023-07/2023
Algorithm Researcher

- **Hardware-Aware NAS Toolchain Development**: Developed a hardware-aware toolchain for neural network architecture optimization, enhancing computational efficiency and performance.
- **Toolchain Validation**: Validated the toolchain's effectiveness through practical applications in visual tasks

(e.g., image denoising, object detection), analyzing its advantages and limitations for further improvement.

EXTRACURRICULAR ACTIVITIES

- 2023 AAAI Conference on Artificial Intelligence.
- The International Conference on Automated Machine Learning 2023.

LANGUAGE& SKILLS

- IELTS: Overall: 7.0 (Reading: 7.5, Listening: 8.0, Speaking: 5.5, Writing: 6.5)
- Software: Microsoft Office, Git, Vim, Linux, Python, C/C++